Collecting Data from the Web

Basic Statistics Concepts Every Researcher Should Know

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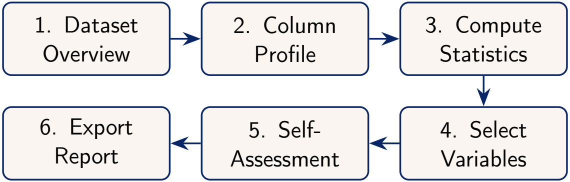
instats Seminar

Friday, May 15, 2026
4:00 PM – 7:00 PM UTC

The Exploration Workflow

Before running complex models, you must understand your data.



- **Descriptive, not Inferential:** We are summarising what is *in* the dataset, not making predictions about the world.
- **Iterative:** Exploration often reveals new cleaning tasks and guides variable selection.

Data Types: How the Computer Sees Your Data

Different types of data require different statistical summaries.

Data Type	**What it is and how we summarise it**
Numeric	Continuous or discrete numbers (e.g., Population, Citation Count). *Summarised by:* Mean, median, standard deviation, histograms.
Categorical	Text labels or groups (e.g., Country, Device Type, Browser). *Summarised by:* Frequencies, percentages, bar charts.
Date/Time	Chronological markers (e.g., 2025-06-13). *Summarised by:* Counts over time (years, months).

Dataset Overview: The Bird's-Eye View

The first step is simply counting the dimensions of your data.

Rows (N)

Units of Observation

- How many records do you have?
- Does this match what you expect from the source?
- Are they individuals, countries, or events?

Columns (P)

Variables

- What attributes were measured?
- High dimensionality (many columns) can complicate analysis.

Missing Cells: The total count of empty values across the entire table gives a rough indicator of dataset quality.

Column Profiling: Unique Values and Missingness

Before computing statistics, we evaluate each column individually.

- **Unique Values (`nunique`):** How many distinct entries exist?
  - If a column has 1 unique value, it provides no information (a constant).
  - If a categorical column has as many unique values as rows, it is likely an ID or a free-text field.
- **Missingness (% Missing):** What proportion of the data is blank?
  - $<$**5%:** Often manageable, but context matters.
  - $>$**20%:** Proceed with caution; the variable may be unreliable.
  - *Caution:* These thresholds are heuristics only. Even a single missing value can bias results if the missingness is systematic. Always ask *why* values are missing.

Descriptive Statistics (Numeric Columns)

We use standard metrics to describe the central tendency and spread.

Central Tendency (Where is the middle?)

- **Mean:** The arithmetic average. Sensitive to extreme values.
- **Median (50th Percentile):** The exact middle value. Robust to extremes.

Dispersion (How spread out is it?)

- **Standard Deviation (σ):** Typical spread around the mean ($\sigma = \sqrt{\text{Variance}}$).
- **Min / Max:** The boundaries of your data.
- **Quartiles (25th, 75th):** Help identify the interquartile range (IQR).

Histograms: Visualising Distributions

A histogram groups numeric data into **bins** to show its shape.

Why look at the shape?

- **Normal (Bell Curve):** Symmetrical. Mean $\approx$ Median.
- **Right-Skewed:** Long tail to the right (e.g., income, citation counts). Mean $>$ Median.
- **Bimodal:** Two peaks. May indicate two distinct groups mixed together.

The Binning Problem:
Changing the number of bins can drastically alter how the distribution looks. Always test different bin sizes (e.g., 10, 20, 50) to find the true shape.

Categorical Columns: Value Counts

For text or grouping variables, we count the frequency of each category.

- **Frequency Tables:** Show the raw count and percentage for each category.
- **Top-N Analysis:** If a column has 1,000 unique categories (e.g., cities), we often only plot the Top 10 or 20 to avoid visual clutter.
- **Bar Charts:** The standard visual for categorical data. Length of the bar represents the frequency.

Note: Always calculate percentages out of the ***non-missing*** *values, unless the missingness itself is a category of interest.*

Data Quality Issues in Scraped Data

Web-scraped data introduces specific quality problems beyond simple missingness.

- **Duplicate Rows:** The same record may appear multiple times due to pagination overlap or repeated API calls. Always deduplicate before analysis.
- **Inconsistent Encodings:** The same entity may appear as `"US"`, `"USA"`, or `"United States"`. Standardise before grouping.
- **Unit Inconsistencies:** Numeric values may mix units (e.g., thousands vs. millions). Check metadata carefully.
- **Sampling / Platform Bias:** Scraped data reflects *what was published on that platform*, not the underlying population. Twitter data reflects Twitter users; OpenAlex reflects indexed publications. This is a fundamental limitation that no cleaning step can fully correct.

Correlation Matrix: Relationships Between Variables

A correlation matrix shows how strongly pairs of numeric columns move together.

Pearson Correlation Coefficient (r):

- **+1.0:** Perfect positive relationship (both go up together).
- **0.0:** No linear relationship.
- **−1.0:** Perfect negative relationship (one goes up, the other goes down).

Crucial Caveats

- **Linearity:** Pearson only detects straight-line relationships. A perfect U-shape curve will have an r near 0.
- **Correlation $\neq$ Causation:** Correlations may arise from confounding variables or common causes — not a direct link between the two variables.

Outlier Detection: The 3σ Rule

An outlier is a value that is abnormally distant from the rest of the data.

The 3 Standard Deviations (3σ) Rule:

- In a *normal* distribution, 99.7% of data falls within 3 standard deviations of the mean.
- Values beyond $\pm 3\sigma$ are statistically rare.
- *This heuristic is less appropriate for highly skewed distributions* (e.g., citation counts, income), where extreme values are expected.

What to do with them?

- **Data Entry Error?** (e.g., Human age $= 999$) $\rightarrow$ Remove or correct.
- **Genuine Extreme?** (e.g., Wealth of a billionaire) $\rightarrow$ Keep, but consider robust statistics (median instead of mean) or logarithmic transformations.

The Four-Criteria Self-Assessment

Before concluding your analysis, evaluate your dataset against these four pillars:

1. **Representativeness:** Does the sample reflect the population you want to study? (Did API pagination limits cut off half your data?)
2. **Completeness:** Are there systematic missing values? (e.g., Are older records missing more data than newer ones?)
3. **Consistency:** Are values recorded in the same format throughout? (e.g., `"US"`, `"USA"`, `"United States"`)
4. **Validity:** Are values logically coherent and within expected ranges?

Ethics and Open Science

Scraping and API usage come with responsibilities.

- **Terms of Service:** Always check if a site explicitly forbids scraping.
- `robots.txt`: Specifies which paths a crawler may access. Note that `robots.txt` is *advisory*, not necessarily legally binding — but violating it may breach Terms of Service.
- **Rate Limiting:** Do not overwhelm servers. Use delays (`sleep`) in your code.
- **Privacy:** Be cautious when collecting personally identifiable information (PII), even if it is publicly visible.
- **Reproducibility:** Share your scripts (not just your data) so others can replicate your collection process.
- **Attribution:** Always cite the original data source.

Summary: The Three-Day Arc

Day 1: Collection
Identifying sources, GET/POST requests, JSON structure.

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Day 2: Cleaning
Handling missingness, standardising formats, selective flattening.

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Day 3: Exploration
Profiling, descriptive statistics, critical assessment.

Thank you for participating!

EDA and cleaning are iterative — exploration continues throughout modelling.